

HARSHITHA PUTTASWAMY

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PROFESSIONAL EXPERIENCE

Software Developer, Internship

February 2025 – Present

Binghamton University, Department of Computer Science

Binghamton, NY, USA

- Developed a dynamic Fair Allocation and LLM Evaluation Framework using Python, generating over 10,000+ distinct allocation scenarios, evaluating LLMs like Deepseek, Gemma, and Gwen for perceived fairness.
- Enhanced computational throughput by 3x as measured by parallel execution benchmarks, by implementing multi-threaded valuation distribution with robust error handling and caching of subroutines
- Enabled reproducible, scalable experimentation workflows as measured by 0% data loss and 100% reproducibility, by building checkpointing, structured logging, and modular interfaces across modules.
- Created evaluation and analysis tools for pairwise allocation comparisons, generating dynamic heatmaps and histograms for property distributions, enhancing data-driven decision-making.

Research Assistant, Internship

January 2024 – December 2024

Binghamton University, Department of Computer Science

Binghamton, NY, USA

- Developed a scalable iterative voting simulation framework supporting 8+ complex voting algorithms, improving system modularity by integrating Gurobi-based ILP solvers for optimal committee selection.
- Engineered a multi-agent simulation system that processed 50,000+ iterations per run, increasing computational efficiency by automating agent-voter interactions using epsilon-greedy Multi-Armed Bandit models.
- Automated performance benchmarking across voting rules, reducing manual experiment overhead by 80% through the implementation of reproducible Python pipelines and statistical analysis modules.

Backend Developer, Internship

March 2022 – December 2022

Fyle Technologies Pvt Ltd

Bangalore, India

- Developed and optimized RESTful APIs using Django, Angular, and PostgreSQL (PSQL), and collaborated with PMs to implement 8 new features that increased market competitiveness and user engagement.
- Spearheaded the implementation of a comprehensive CI/CD pipeline using Pytest and GitHub Actions, independently developing tests for 3 core modules that achieved over 98% code coverage each, significantly improving code quality.
- Authored and delivered session on Pytest, GitHub Actions, and CI principles, empowering team adoption of best practices.

PROJECT EXPERIENCE

Immigration Law Chatbot (Fine-Tuned LLM with RAG on AWS) | AWS, Mistral-7B, RAG May 2025 – June 2025

- Developed and Fine-tuned Mistral-7B-instruct LLM on AWS SageMaker for legal Q&A, leveraging a 9,000+ pair custom dataset.
- Architected and deployed a production-ready, scalable LLM inference system on AWS Lambda and API Gateway, reducing hallucination and ensuring answer relevance by integrating Retrieval-Augmented Generation via Amazon OpenSearch.
- Engineered robust data pipelines for LLM training and RAG knowledge base, ensuring reliable and seamless ingestion and retrieval of large datasets and source documents from AWS S3.

Autokube - Kubernetes Cluster Automation Platform | DevOps, CI/CD

August 2021 – March 2022

- Automated Kubernetes cluster creation and modification, automating DevOps workflows for JavaScript and Python-based applications, reducing deployment time by over 80%.
- Orchestrated the dynamic generation of Dockerfiles and Kubernetes manifests based on user input, ensuring consistent and reproducible deployments, leading to a 40% reduction in infrastructure setup time.
- Developed robust APIs using Express.js, designed reusable components using React.js, and MongoDB for user data storage, enabling deployments with just 3 clicks.

EDUCATION

Master of Science in Computer Science

January 2023 – December 2024

Binghamton University, SUNY (State University of New York)

Binghamton, NY, USA

Relevant Coursework: Deep Learning, High Performance Computing, Computer Architecture, Design and Analysis of Algorithm

TECHNICAL SKILLS

Programming Languages: C, CPP, Python, Java, JavaScript, TypeScript, MongoDB, PostgreSQL, MySQL, Firebase

Frameworks and Libraries: TensorFlow, Keras, PyTorch, CUDA, OpenMP, MPI (Familiarity via HPC coursework)

Cloud and Tools: AWS, Git, Linux, Docker, Kubernetes, Github Actions, Pytest

Machine Learning and AI: Deep Learning, Reinforcement Learning, NLP, LLM, RAG

Core Computer Science: Data Structures, Algorithm Design & Analysis, Distributed Systems, Computer Architecture

CERTIFICATIONS

AWS Certified Machine Learning Engineer - Associate

- Validates expertise in building, deploying, and managing machine learning solutions on AWS